

Homework 2

Due Wednesday, February 5

- (10 pts.) The group S_3 discussed in Homework 1 has a subgroup, called C_3 (also Z_3), which consists of three elements: e , r , and r^2 . The group is Abelian, so each element is a separate conjugacy class. The character table for the irreps of C_3 is as follows:

	e	r	r^2
A	1	1	1
C	1	ϵ	ϵ^2
C^*	1	ϵ^2	ϵ

where $\epsilon = e^{2\pi i/3}$.

- Consider the 3-dimensional representation (call it $\mathcal{D}$) described in HW 1 Prob. 2, now as a representation of C_3 . The matrix corresponding to r in it is

$$D(r) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Compute the characters of this representation.

- Decompose $\mathcal{D}$ into a direct sum of the irreps of C_3 . (You do not have to construct the projectors on the corresponding subspaces.)

- (5 pts.) Problem 2.A from the textbook.
- (5 pts.) (a) Prove the repeated commutator formula

$$e^{\tau A} B e^{-\tau A} = \sum_{n=0}^{\infty} \frac{\tau^n}{n!} [A, [A, \dots [A, B]]],$$

where A and B are square matrices of the same size and τ is a parameter; in the n -th term on the right-hand side, A occurs n times. *Hint:* Consider the n -th derivative with respect to τ .

- Use the formula from (a) to solve Problem 2.B from the textbook.